

ANTOINE PIERRARD

Junior R&D Engineer

Machine Learning • MedTech • Embedded Systems • Computer Vision

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Junior R&D Engineer with expertise in robotics, electronics, and cutting-edge medical devices. Driven to contribute to medical innovation, particularly in neuroscience and oncology, with the conviction that the medicine of tomorrow cannot be built without engineers

RESEARCH & ENGINEERING EXPERIENCE

BOCCONI UNIVERSITY / POLITECNICO DI MILANO

2026

Computational Neuroscience Research Intern - Master's Thesis | Milan, Italy

Supervised by *Prof. Alessandro Sanzeni*

- Developed latent-space machine learning models to predict connectivity in the visual cortex from large-scale MICrONS connectome data
- Analyzed the learned representations to identify the biological information they encode and uncover higher-order connectivity principles

Python • PyTorch • MICrONS/CAVE • NumPy • pandas • scikit-learn

INRIA PARIS-SACLAY / ÉCOLE POLYTECHNIQUE - BOOST TEAM

2025

Machine Learning Research Intern | Palaiseau, France

- Developed and implemented an interpretable machine-learning model leveraging multimodal physiological data (ECG, EEG) to differentiate mental and physical stress.
- Represented physiological signals as visibility graphs and optimized Explainable Boosting Machine models with Optuna.

Python • EEG/ECG • Visibility Graphs • Explainable Boosting Machine • Optuna

INRIA NANCY

2024-2025

Project Lead | Nancy, France

- Developed a camera network with our own patient re-identification and tracking algorithm in an emergency department, using computer vision to map patients, predict wait times and optimize staff intervention.

OpenCV • YOLO • MediaPipe • Python

TÉLÉCOM PHYSIQUE STRASBOURG

2023-2024

MicroMix - Embedded Systems Project

- Designed an embedded system for high-throughput cell sorting with a LabVIEW GUI, optimizing an Arduino microcontroller through low-level register programming, achieving a 10× speed increase

LabVIEW • Arduino • C • Embedded Systems

EDUCATION

Télécom Physique Strasbourg

2023-2026

Institut Mines-Télécom • Engineering Degree (MSc equivalent) - MedTech
R&D-oriented program: haptic robotics, medical imaging, computer vision, signal processing, microelectronics and microcontroller programming.

Polytechnique Montréal

2025-2026

Academic Exchange - Electrical Engineering & Robotics
Electrical engineering, electronics and robotics coursework.

Lycée Blaise Pascal, Clermont-Ferrand

2020-2023

CPGE MPSI/MP

Highly competitive preparatory program in advanced mathematics and physics.

LANGUAGES

French Native • English C1 • German B2

TECHNICAL SKILLS

Programming: Python, C, Java, Kotlin, OCaml, VHDL, Verilog

Machine Learning: PyTorch, TensorFlow, scikit-learn, Optuna, NumPy, pandas

Embedded Systems & Robotics: Arduino, STM32, FPGA, LabVIEW, ROS

Signal Processing: MATLAB, FT/FFT, analog & digital filters, EEG/ECG

Computer Vision & Imaging: OpenCV, ITK-SNAP, FIJI, YOLO, MediaPipe

Prototyping: SolidWorks, COMSOL, HyperMesh, KiCad, 3D printing, laser cutting

INTERESTS

Tennis (French ranking 15/2) • Running • Alpine skiing • Chess • Drama